Mathematical
Inequalities
& Applications
Volume 2, Number 1 (1999), 19–24

EXTENSIONS OF THE GEOMETRIC–ARITHMETIC MEANS INEQUALITY TO A DISC OF THE COMPLEX PLANE

RICHARD FOURNIER

Abstract. For complex numbers z_j with $|z_j - 1| \leqslant r$, $r < 1$, we consider the arithmetic mean $A_n := (1/n)\sum_{j=1}^{n} z_j$ and the geometric mean $G_n := \prod_{j=1}^{n} z_j^{1/n}$ and prove, amongst other results, that

$$\frac{1-r^2}{|2-A_n|} \leqslant |G_n| \leqslant \frac{|A_n|}{\sqrt{1-r^2}}.$$

Mathematics subject classification (1991): 26D15, 30C45.
Key words and phrases: geometric and arithmetic means in the complex plane, starlike univalent functions.

REFERENCES

[1] E. F. BECKENBACH AND R. BELLMAN, *Inequalities*, Springer-Verlag, Berlin, 1971.
[2] P. L. DUREN, *Univalent functions*, Springer-Verlag, New-York, 1983.
[3] G. H. HARDY, J. E. LITTLEWOOD, AND G. PÓLYA, *Inequalities*, Cambridge University Press, Cambridge, 1952.
[4] D. S. MITRINOVIĆ, *Analytic inequalities*, Springer-Verlag, Berlin, 1970.
[5] ST. RUSCHEWEYH, *Convolutions in geometric function theory*, Les Presses de l'Université de Montréal, Montréal, 1982.
[6] T. B. SHEIL-SMALL, *Convolution of convex functions*, J. London Math. Soc. **2** (1969), 483– 492.

© ELEMENT, Zagreb
Paper MIA-02-03